Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We respond below to R1–ZQwd, R2–jAyD, and R3–Fcwd. The new evidence directly addresses the requested progress baselines, optimization ambiguity, mask scope, component ablations, and evaluation protocol. We will revise the paper to narrow the claim to an online, tactile-supervised short-horizon contact prior that reduces pre-contact stall.

[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning? We agree that this comparison is necessary. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. VPCE obtains AUC/F1 of $.963/.759$, $.885/.705$, and $.946/.735$ on SWIPE/WIPE/INSERT, versus $.853/.597$, $.743/.406$, and $.808/.539$ for progress and $.670/.000$, $.666/.000$, and $.684/.000$ for phase. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

TABLE I
CONTACT PREDICTION AUC/F1 USING PHASE, PROGRESS, OR VPCE.

Task	Phase-4	Progress-20	VPCE
SWIPE	$.670/.000$	$.853/.597$	$\mathbf{.963/.759}$
WIPE	$.666/.000$	$.743/.406$	$\mathbf{.885/.705}$
INSERT	$.684/.000$	$.808/.539$	$\mathbf{.946/.735}$

This is a predictor-level comparison, not a full reproduction of every hierarchical policy. We will add Hierarchical Diffusion Policy and CompILE, include these controlled baselines, and avoid claiming superiority over all hierarchical methods. Unlike a discrete phase switch, VPCE continuously conditions the same action generator with an online, state-dependent 15-step contact horizon.

[R1, R3] Does the teacher transfer, and how much target tactile data is required? We do not claim universal zero-shot transfer. The source teacher gives AUC/F1 $.500/.095$ on board and $.662/.666$ on card; target adaptation raises board to $.897/.750$. The current target teacher uses approximately 60 demonstrations with tactile recordings per task. Tactile signals generate contact labels automatically, without manual frame annotation, but collecting target-domain tactile demonstrations remains a limitation. We will report these numbers and state this transfer boundary explicitly.

[R2] Learned loss weights collapse toward the lower bound. We agree. Optimizing γ and δ through the same weighted denoising objective creates a direct incentive to reduce the weight. Checkpoint inspection shows weights of only 1.000–1.005; the lower bound prevents down-weighting below ordinary DP loss, but the learned component becomes ineffective. We therefore replace it with the fixed schedule $w(c) = \mathrm{clip}(1 + c, 1, 2)$. This α=1 version obtains $8/10$ on SWIPE and $7/10$ on WIPE. Together with the submitted no-weighting result ($8/10$ on SWIPE), this shows that the main contact-anticipation result does not depend on the problematic learned weight. We will present the fixed results as small-N functionality checks, not as a claim that weighting alone improves success.

[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth? The reviewer is correct that the paper's notation and original post-encoder routing conflated proprioception, gripper width, and tactile features. Although input-level dropout targeted tactile keys, the post-encoder mask could zero the entire low-dimensional tail. We corrected the routing to always retain robot state and gripper width and mask only tactile/contact features. This controlled variant obtains $13/20$ on SWIPE and produces more coherent free-space approach motion. Action-chunk prediction and receding-horizon execution provide additional temporal continuity. We will revise Eq. (9) to distinguish f_{prop} from f_{tac}.

[AC, R3] Which components matter, and why retain gating? We agree that the main-text ablation was insufficient. The submitted SWIPE pilot gives: full $9/10$, no horizon $2/10$, horizon only $0/10$, no gating $9/10$, no loss weighting $8/10$, and no masking $0/10$. Therefore, the evidence does *not* support gating as necessary. Horizon conditioning and contact-adaptive masking are the main mechanisms; loss weighting is secondary, and gating is an optional/task-dependent modulation. To answer R2's visualization request, checkpoint probing shows the mean visual gate decreasing from .935 to .313 and the low-dimensional gate increasing from .937 to 2.874 as mean contact probability moves from 0 to 1. We will include this trace as diagnostic evidence, move the ablation table into the main text, and revise the method and contribution statements accordingly.

[R2] Could operator-held props assist the policy? The reader and foam were held only to maintain a predefined height and orientation in the reachable workspace; the holder did not track the end effector, adjust the object online, or assist alignment after rollout start. All methods used the same placement, success criterion, and 60-s timeout. We expanded the original VPCE-DP evaluation to 30 trials per task: SWIPE 28/30 (93.3%, Wilson 95% CI $[78.7, 98.2]$%), WIPE 22/30 (73.3%, $[55.6, 85.8]$%), and INSERT 18/30 (60.0%, $[42.3, 75.4]$%). These results are consistent with the submission, but we agree that operator compliance remains a confound and will not use this protocol to claim fixed-environment robustness.

The height constraint also clarifies the method's scope: VPCE predicts only 15 future steps and is not a long-range approach planner. Starting far outside this horizon can leave contact probability low and reintroduce stalling. We will add this limitation and investigate a height-matched rigid fixture plus an external-view camera or coarse approach controller that first reaches the interaction region before wrist-based VPCE regulates imminent contact.

[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled? VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame z-force proxy. A threshold of 0.095 is reused for most tasks; where sensor scale differs, it is selected once from the training force-change distribution and then fixed. We acknowledge this calibration requirement. A $0.6\times$–$1.4\times$ threshold sweep changes the 15-step positive-label rate smoothly (SWIPE $.296 \rightarrow .217$, WIPE $.582 \rightarrow .379$, INSERT $.390 \rightarrow .290$), rather than creating labels only at one tuned value; this is label sensitivity, not yet a policy-success robustness claim. Contacts with changing normals can instead use $\|\Delta\mathbf{F}\|_2$, local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning? We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.